Final Exam Covid Data

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Overview

This report pulls the daily COVID-19 time series data and builds a few visualizations and a simple linear model looking at the relationship between how widely a state was infected and how deadly the outbreak was there.

Importing the Data

```
url_in <- "https://raw.githubusercontent.com/CSSEGISandData/COVID-19/refs/heads/master/csse_covid_19_da

file_names <- c("time_series_covid19_confirmed_US.csv",
                "time_series_covid19_confirmed_global.csv",
                "time_series_covid19_deaths_US.csv",
                "time_series_covid19_deaths_global.csv")

urls <- str_c(url_in, file_names)

us_cases      <- read_csv(urls[1])
global_cases  <- read_csv(urls[2])
us_deaths     <- read_csv(urls[3])
global_deaths <- read_csv(urls[4])
```

Tidying the Global Data

Each global file has one row per country and a separate column for every date. The block below pivots those date columns into a single `date` column with a `cases`/`deaths` value column, drops `Lat`/`Long`joins the cases and deaths tables together, converts `date` from text to an actual date, and renames the `Province/State` and `Country/Region` columns to underscore-separated names.

```
global_cases <- global_cases %>%
  pivot_longer(cols = -c(`Province/State`, `Country/Region`, Lat, Long),
               names_to = "date", values_to = "cases") %>%
  select(-c(Lat, Long))

global_deaths <- global_deaths %>%
  pivot_longer(cols = -c(`Province/State`, `Country/Region`, Lat, Long),
               names_to = "date", values_to = "deaths") %>%
  select(-c(Lat, Long))
```

```r
global <- global_cases %>%
  full_join(global_deaths,
            by = c("Province/State", "Country/Region", "date")) %>%
  rename(Country_Region = `Country/Region`,
         Province_State = `Province/State`) %>%
  mutate(date = mdy(date)) %>%
  filter(cases > 0)
```

Adding Population Data to the Global Data

The time series files don't include population, so this brings it in. It also combines the state or province with the country in its own column.

```r
global <- global %>%
  unite("Combined_Key", c(Province_State, Country_Region),
        sep = ", ", na.rm = TRUE, remove = FALSE)

uid_lookup_url <- "https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/

uid <- read_csv(uid_lookup_url, show_col_types = FALSE) %>%
  select(-c(Lat, Long_, Combined_Key, code3, iso2, iso3, Admin2))

global <- global %>%
  left_join(uid, by = c("Province_State", "Country_Region")) %>%
  select(-c(UID, FIPS)) %>%
  select(Province_State, Country_Region, date, cases, deaths,
         Population, Combined_Key)
```

Tidying the US Data

The US files had many unneeded rows and columns. This code cleans them up

```r
us_cases <- us_cases %>%
  pivot_longer(cols = -(UID:Combined_Key),
               names_to = "date", values_to = "cases") %>%
  select(Admin2:cases) %>%
  mutate(date = mdy(date)) %>%
  select(-c(Lat, Long_))

us_deaths <- us_deaths %>%
  pivot_longer(cols = -(UID:Population),
               names_to = "date", values_to = "deaths") %>%
  select(Admin2:deaths) %>%
  mutate(date = mdy(date)) %>%
  select(-c(Lat, Long_))

us <- us_cases %>%
  full_join(us_deaths,
            by = c("Admin2", "Province_State", "Country_Region",
                   "Combined_Key", "date"))
```

Aggregating to State and National Level

Next one gives one row per state and cleans up the totalsa

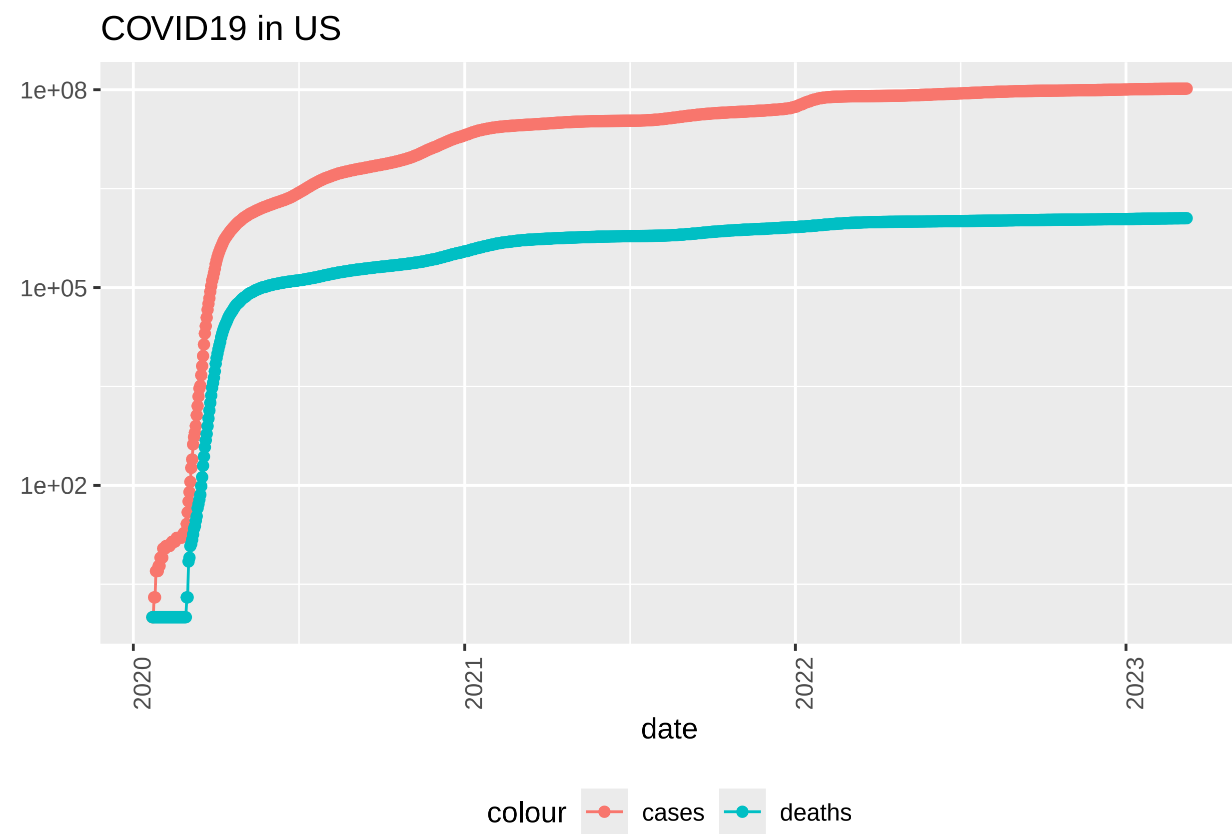
```
us_by_state <- us %>%
  group_by(Province_State, Country_Region, date) %>%
  summarize(cases = sum(cases), deaths = sum(deaths),
            Population = sum(Population), .groups = "drop") %>%
  mutate(deaths_per_mill = deaths * 1000000 / Population) %>%
  select(Province_State, Country_Region, date,
         cases, deaths, deaths_per_mill, Population)

us_totals <- us_by_state %>%
  group_by(Country_Region, date) %>%
  summarize(cases = sum(cases), deaths = sum(deaths),
            Population = sum(Population), .groups = "drop") %>%
  mutate(deaths_per_mill = deaths * 1000000 / Population) %>%
  select(Country_Region, date, cases, deaths, deaths_per_mill, Population)
```

Visualization 1: Cumulative Cases and Deaths in the US

This plot shows the running total of confirmed cases and deaths for the country over time.

```
us_totals %>%
  filter(cases > 0) %>%
  ggplot(aes(x = date, y = cases)) +
  geom_line(aes(color = "cases")) +
  geom_point(aes(color = "cases")) +
  geom_line(aes(y = deaths, color = "deaths")) +
  geom_point(aes(y = deaths, color = "deaths")) +
  scale_y_log10() +
  theme(legend.position = "bottom",
        axis.text.x = element_text(angle = 90)) +
  labs(title = "COVID19 in US", y = NULL)
```

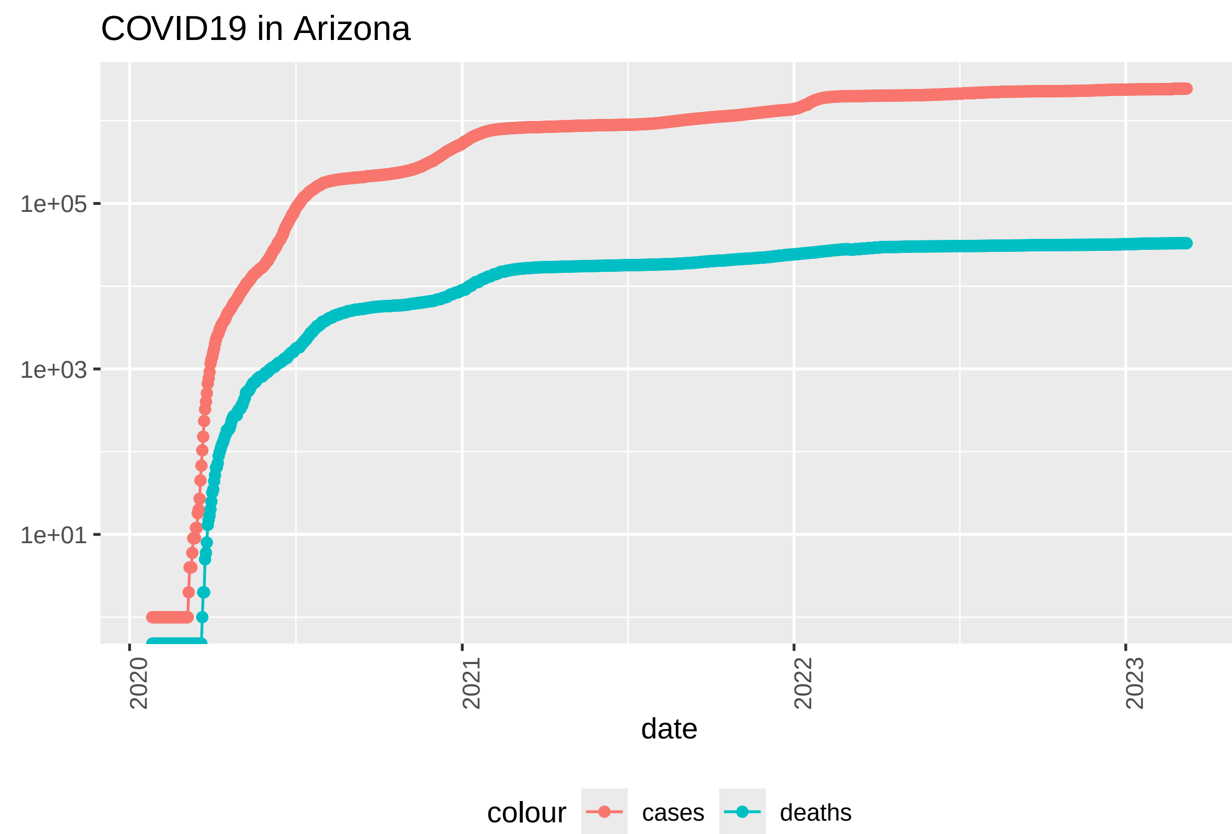


Visualization 2: Cumulative Cases and Deaths for a Single State

Same idea as above, but filtered down to just Arizona

```
state <- "Arizona"

us_by_state %>%
  filter(Province_State == state, cases > 0) %>%
  ggplot(aes(x = date, y = cases)) +
  geom_line(aes(color = "cases")) +
  geom_point(aes(color = "cases")) +
  geom_line(aes(y = deaths, color = "deaths")) +
  geom_point(aes(y = deaths, color = "deaths")) +
  scale_y_log10() +
  theme(legend.position = "bottom",
        axis.text.x = element_text(angle = 90)) +
  labs(title = str_c("COVID19 in ", state), y = NULL)
```



Daily New Cases and Deaths

The columns built so far are cumulative totals,which are useful for seeing overall scale but this next code allows to see the change day to day.

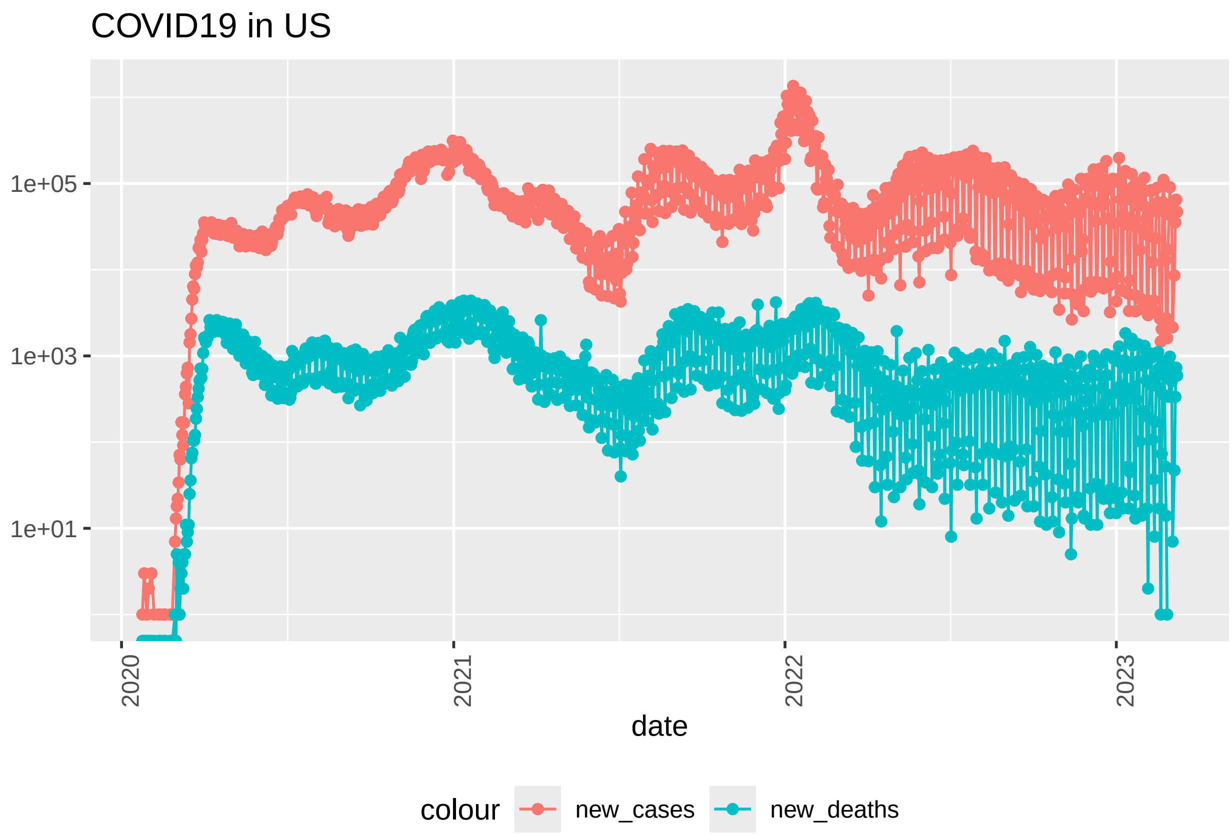
```
us_by_state <- us_by_state %>%
  group_by(Province_State) %>%
  mutate(new_cases = cases - lag(cases),
         new_deaths = deaths - lag(deaths)) %>%
  ungroup()

us_totals <- us_totals %>%
  mutate(new_cases = cases - lag(cases),
         new_deaths = deaths - lag(deaths))
```

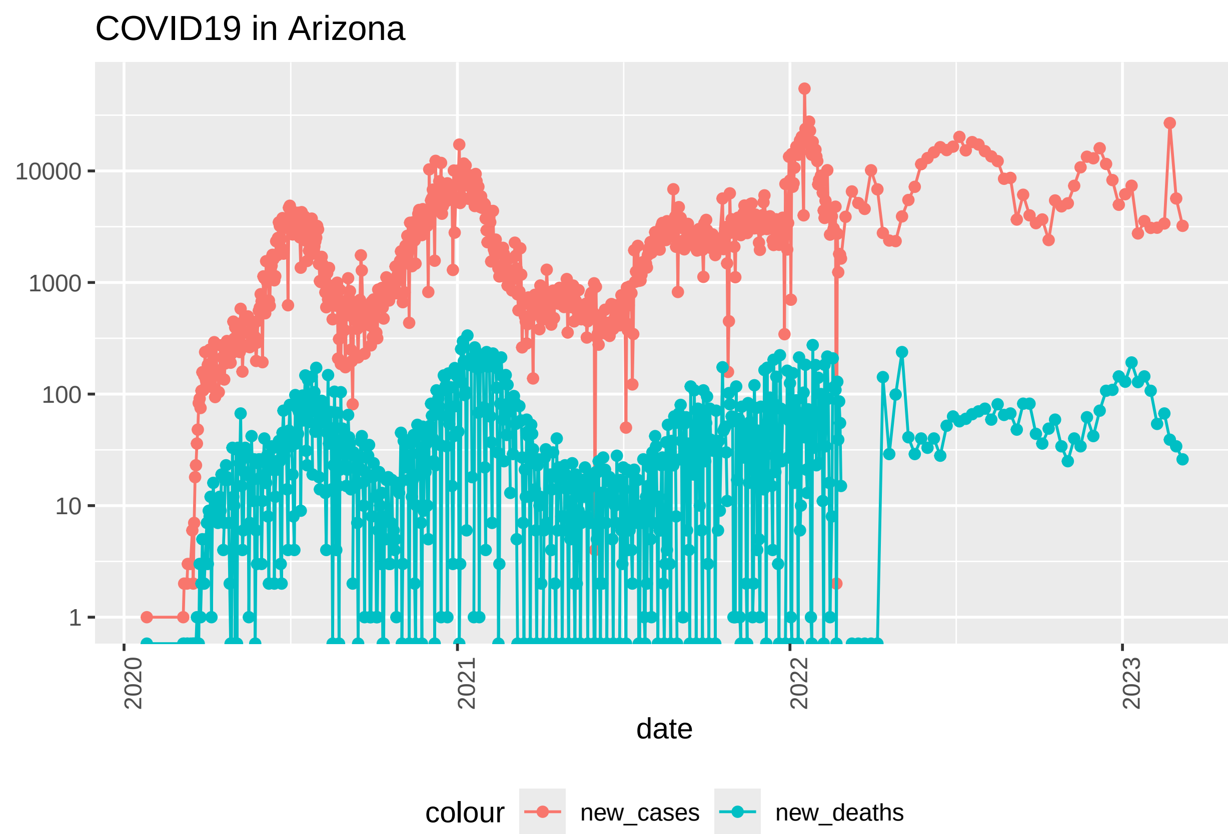
Visualization 3: Daily New Cases and Deaths

Same log-scale line/point pattern as before, but now showing the daily change rather than the running total.When seeing this you can see that the values begin to decrease as the dates continue forward and historically this makes sense as there was more of a handle on vaccines and quarantine practices. Also you can see just how much more spread out it is in the newer dates compared to the height of Covid in 2020/2021.

```r
us_totals %>%
  filter(new_cases > 0) %>%
  ggplot(aes(x = date, y = new_cases)) +
  geom_line(aes(color = "new_cases")) +
  geom_point(aes(color = "new_cases")) +
  geom_line(aes(y = new_deaths, color = "new_deaths")) +
  geom_point(aes(y = new_deaths, color = "new_deaths")) +
  scale_y_log10() +
  theme(legend.position = "bottom",
        axis.text.x = element_text(angle = 90)) +
  labs(title = "COVID19 in US", y = NULL)
```




```r
us_by_state %>%
  filter(Province_State == state, new_cases > 0) %>%
  ggplot(aes(x = date, y = new_cases)) +
  geom_line(aes(color = "new_cases")) +
  geom_point(aes(color = "new_cases")) +
  geom_line(aes(y = new_deaths, color = "new_deaths")) +
  geom_point(aes(y = new_deaths, color = "new_deaths")) +
  scale_y_log10() +
  theme(legend.position = "bottom",
        axis.text.x = element_text(angle = 90)) +
  labs(title = str_c("COVID19 in ", state), y = NULL)
```



State-Level Summary: Rates per Thousand Residents

This collapses each state down to a single row with its maximum values and makes them in rate per thousand.

```
us_state_totals <- us_by_state %>%
  group_by(Province_State) %>%
  summarize(deaths = max(deaths), cases = max(cases),
            population = max(Population),
            cases_per_thou = 1000 * cases / population,
            deaths_per_thou = 1000 * deaths / population) %>%
  filter(cases > 0, population > 0)
```

Exploring the Best and Worst States by Death Rate

A quick look at which states had the lowest and highest deaths per thousand.When looking at this you can see that Arizona had the highest rate be thousand and American Samoa had the least.

```
us_state_totals %>%
  slice_min(deaths_per_thou, n = 10) %>%
  select(deaths_per_thou, cases_per_thou, everything())
```

```
## # A tibble: 10 x 6
##    deaths_per_thou cases_per_thou Province_State        deaths   cases population
```

```
##              <dbl>          <dbl> <chr>                   <dbl>   <dbl>      <dbl>
##  1          0.611          150. American Samoa             34 8.32e3      55641
##  2          0.744          248. Northern Mariana Isl~      41 1.37e4      55144
##  3          1.21           231. Virgin Islands            130 2.48e4     107268
##  4          1.30           269. Hawaii                   1841 3.81e5    1415872
##  5          1.49           245. Vermont                   929 1.53e5     623989
##  6          1.55           293. Puerto Rico              5823 1.10e6    3754939
##  7          1.65           340. Utah                     5298 1.09e6    3205958
##  8          2.01           415. Alaska                   1486 3.08e5     740995
##  9          2.03           252. District of Columbia     1432 1.78e5     705749
## 10          2.06           253. Washington              15683 1.93e6    7614893
```

```
us_state_totals %>%
  slice_max(deaths_per_thou, n = 10) %>%
  select(deaths_per_thou, cases_per_thou, everything())
```

```
## # A tibble: 10 x 6
##    deaths_per_thou cases_per_thou Province_State deaths   cases population
##              <dbl>          <dbl> <chr>           <dbl>   <dbl>      <dbl>
##  1            4.55           336. Arizona         33102 2443514    7278717
##  2            4.54           326. Oklahoma        17972 1290929    3956971
##  3            4.49           333. Mississippi     13370  990756    2976149
##  4            4.44           359. West Virginia    7960  642760    1792147
##  5            4.32           320. New Mexico       9061  670929    2096829
##  6            4.31           334. Arkansas        13020 1006883    3017804
##  7            4.29           335. Alabama         21032 1644533    4903185
##  8            4.28           368. Tennessee       29263 2515130    6829174
##  9            4.23           307. Michigan        42205 3064125    9986857
## 10            4.06           385. Kentucky        18130 1718471    4467673
```

Linear Model: Deaths per Thousand vs. Cases per Thousand

This model asks a simple question: across states, does a higher case rate predict a higher death rate?

```
mod <- lm(deaths_per_thou ~ cases_per_thou, data = us_state_totals)
summary(mod)
```

```
##
## Call:
## lm(formula = deaths_per_thou ~ cases_per_thou, data = us_state_totals)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.3352 -0.5978  0.1491  0.6535  1.2086
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.36167    0.72480  -0.499     0.62
## cases_per_thou  0.01133    0.00232   4.881 9.76e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

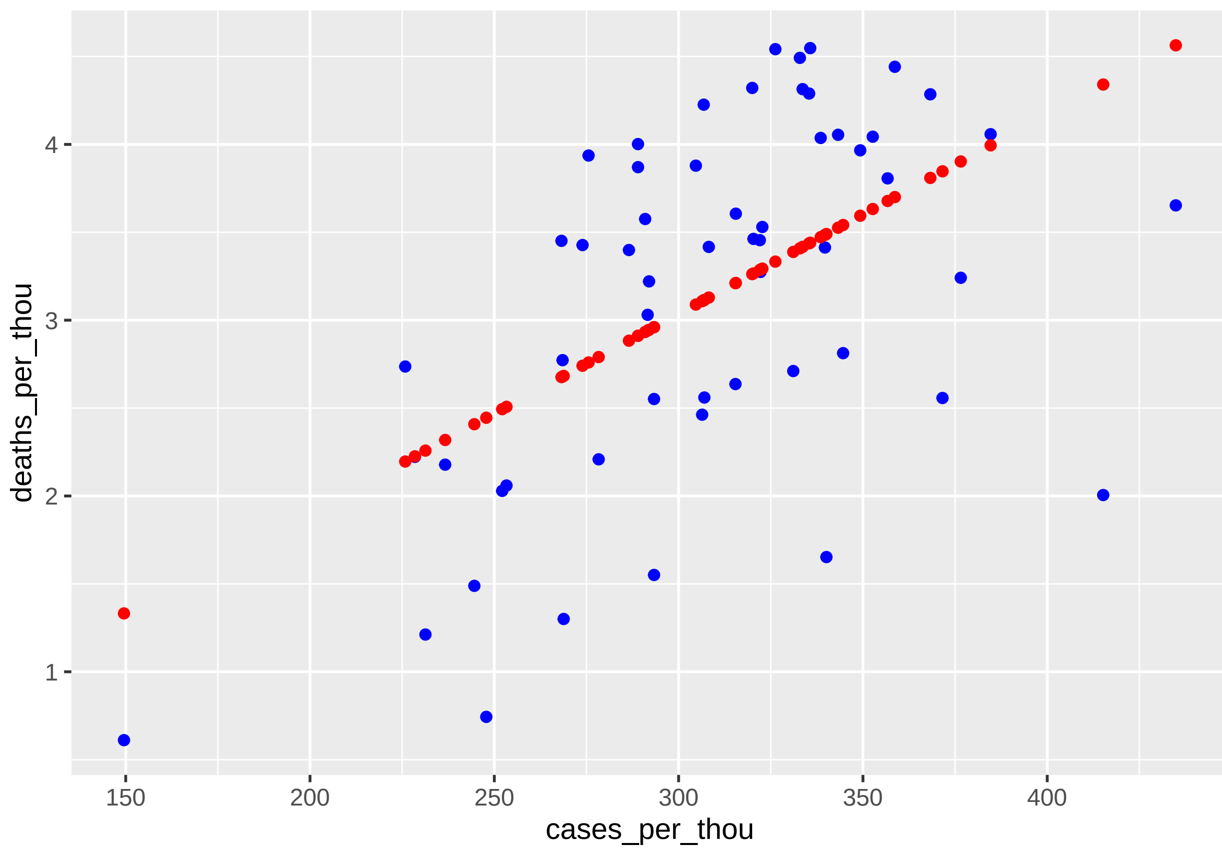
```
## Residual standard error: 0.8615 on 54 degrees of freedom
## Multiple R-squared:  0.3061, Adjusted R-squared:  0.2933
## F-statistic: 23.82 on 1 and 54 DF,  p-value: 9.763e-06
```

Visualization 4: Actual vs. Predicted Deaths per Thousand

This plots each state's actual death rate (blue) against the death rate the model predicts from its case rate alone (red). Where the red and blue points for a state are far apart, that state's actual outcome deviated a lot from what its case rate alone would suggest.

```r
us_tot_w_pred <- us_state_totals %>%
  mutate(pred = predict(mod))

us_tot_w_pred %>%
  ggplot() +
  geom_point(aes(x = cases_per_thou, y = deaths_per_thou), color = "blue") +
  geom_point(aes(x = cases_per_thou, y = pred), color = "red")
```



Bias

There is bias in the variables we have chosen and how we even cleaned the data. It is important to see that we are decding all of the variables and what we are actually looking at in each visualization. This was all set up as our own choice to what we wanted to look at. These numbers are also a little outdated because the newest date was in 2023. This means that there are 3 years of missing data.